

Guardrail Dataset Construction and Analysis

1 Dataset construction and characterization

To evaluate the effectiveness of the proposed guardrail framework under adversarial prompting conditions, we constructed a curated dataset consisting of benign prompts, jailbreak attacks, and harmful intent prompts. The dataset was assembled from multiple publicly available prompt repositories and adversarial prompt datasets.

All prompts were standardized through a preprocessing pipeline involving normalization, duplicate removal, and validation checks to ensure data consistency. Each prompt was annotated with a semantic label indicating whether the prompt represents benign user intent, a jailbreak attempt designed to bypass safety restrictions, or a harmful request explicitly asking for disallowed content.

In addition to the primary label, jailbreak prompts were further categorized according to their underlying attack strategy. This taxonomy allows fine-grained evaluation of guardrail robustness across multiple adversarial prompting techniques.

1.1 Dataset composition

The final dataset contains a total of 1,500 prompts spanning three primary classes. The dataset was intentionally constructed with a slightly larger proportion of adversarial prompts to enable robust evaluation of guardrail systems under challenging scenarios.

Metric	Value
Total prompts	1500
Benign prompts	670
Jailbreak prompts	455
Harmful prompts	375
Training split	1045
Validation split	222
Test split	233

Table 1: Overall dataset composition and dataset splits used for guardrail training and evaluation.

The class distribution enables evaluation of both false-positive and false-negative guardrail behaviors. Benign prompts assess over-blocking, while jailbreak and harmful prompts enable systematic testing of safety

violations.

1.2 Jailbreak attack taxonomy

Adversarial prompts were categorized according to their attack strategy to evaluate guardrail performance across different jailbreak mechanisms.

Attack strategy	Count
Role-play attack	205
Instruction override	124
Multi-step attack	84
Prompt injection	41
Obfuscation	1

Table 2: Distribution of jailbreak attack strategies in the dataset.

Role-play attacks constitute the largest category. These attacks instruct the model to simulate fictional scenarios where safety restrictions are ignored. Instruction override attacks attempt to explicitly negate system instructions, while multi-step attacks embed malicious instructions within reasoning chains.

The entropy of the jailbreak attack distribution is calculated as:

$$H = 1.81bits$$

indicating moderate diversity in adversarial attack strategies.

1.3 Prompt length characteristics

Prompt length statistics provide insight into structural differences between benign and adversarial prompts.

Statistic	Value
Minimum length	15
Maximum length	2490
Mean length	464.18
Median length	92
90th percentile	1534.1
95th percentile	1874.2

Table 3: Global prompt length statistics across the dataset.

The dataset exhibits a long-tail distribution in prompt length. While benign prompts are typically short queries, adversarial prompts often contain extended contextual instructions intended to manipulate model behavior.

Class	Count	Mean	Median	P90
Benign	670	165.88	58	422
Jailbreak	455	1127.17	1109	2075.8
Harmful	375	192.71	89	544.6

Table 4: Prompt length statistics across dataset classes.

Jailbreak prompts exhibit significantly longer average lengths compared with benign prompts. This pattern reflects the use of complex adversarial instructions designed to bypass guardrail filters.

1.4 Dataset balance and split integrity

Maintaining consistent class distributions across dataset splits is essential for reliable evaluation.

Split	Benign	Jailbreak	Harmful
Train	0.4478	0.3014	0.2507
Validation	0.4505	0.2973	0.2523
Test	0.4378	0.3176	0.2446

Table 5: Class proportions across dataset partitions.

Distribution drift across splits remains below approximately 1.5% for all classes, indicating stable dataset construction and minimal sampling bias.

1.5 Source datasets

Prompts were collected from several public datasets covering different domains and adversarial behaviors.

Source dataset	Count
SQuAD v2 question dataset	281
Alpaca instruction dataset	280
TrustAIRLab benign prompts	102
TrustAIRLab jailbreak dataset	328
ToxicChat harmful prompts	260

Table 6: Primary datasets used for prompt collection.

Combining multiple prompt datasets improves topic diversity, linguistic variation, and adversarial coverage.

1.6 Key observations

Several insights emerge from the dataset analysis:

- The dataset includes a substantial proportion of adversarial prompts, enabling rigorous evaluation of jailbreak detection systems.
- Benign prompts remain well represented to assess false-positive guardrail behavior.
- Jailbreak prompts are significantly longer than benign prompts, reflecting complex adversarial instructions.
- Class proportions remain stable across dataset splits, ensuring reliable evaluation.
- Multiple jailbreak attack strategies are represented, enabling evaluation across diverse adversarial behaviors.

1.7 Data availability

The processed dataset and accompanying metadata are stored in structured JSON and CSV formats. The dataset release includes the complete evaluation dataset, dataset splits, statistical summaries, and metadata describing dataset construction procedures. These artifacts are designed to support reproducible evaluation of guardrail systems.